$$f = \text{edge}/\text{odds}$$

F is the percentage of your bankroll you bet. Say you can bet on Big Brown at 5–1 odds at the Kentucky Derby, meaning, if you bet $1, you would stand to win $5 if Big Brown wins. (Plus, you'd get your $1 back.) Odds = 5.

What about your edge? Your inside tip says Big Brown has a one-in-three chance of winning. That means a $1 bet gives you a one-third chance of ending up with $6 ($5 plus your initial $1 bet). On average, such $1 bets are worth $2 — for a net profit of $1. Your edge is your profit divided by the size of your wager, in this case, $1. Edge = 1.

Plug it all into the formula, and Kelly says you should bet 20 percent of your bankroll on Big Brown.

If you don't get the math, don't worry about it. The aim of the formula is to find the optimal amount to bet. And the rough answer is this: when you have a good thing, you bet big.

As you might imagine, this is useful for investors because they too face a question: how much do I put in any one stock?

Kelly's formula gives you an objective way to think about it. But it has quirks. For one thing, the formula is greedy. "It perpetually takes risks in order to achieve ever-higher peaks of wealth," as Poundstone writes. It is for making the most the fastest, but that goal is not for everyone.

Yet it is also conservative in that it prevents ruin. It has, as one professor put it, an "automatically built-in … airtight survival motive." Even so, it produces large swings in your bankroll. As you can see from our Big Brown example, if you lost — and you would have — you lost 20 percent of your bankroll. This has led some to try to smooth the ride a bit by taking a

"half-Kelly." In other words, if the formula says you put 20 percent of your account in one stock, you put half that amount, or 10 percent.

I favor the half-Kelly because it cuts the volatility drastically without sacrificing much return. Poundstone says a 10 percent return using full Kellys turns into 7.5 percent with half-Kellys. But note this: "The full-Kelly bettor stands a one-third chance of halving her bankroll before she doubles it. The half-Kelly better [sic] has only a one-ninth chance of losing half her money before doubling it."